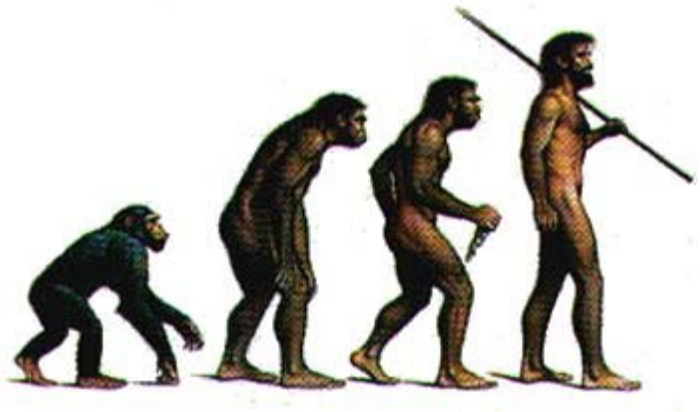
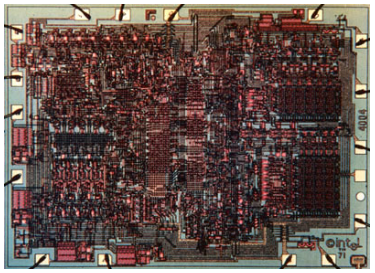
Quantum Computing

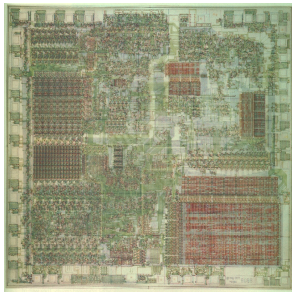
Dave Bacon

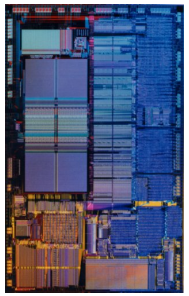








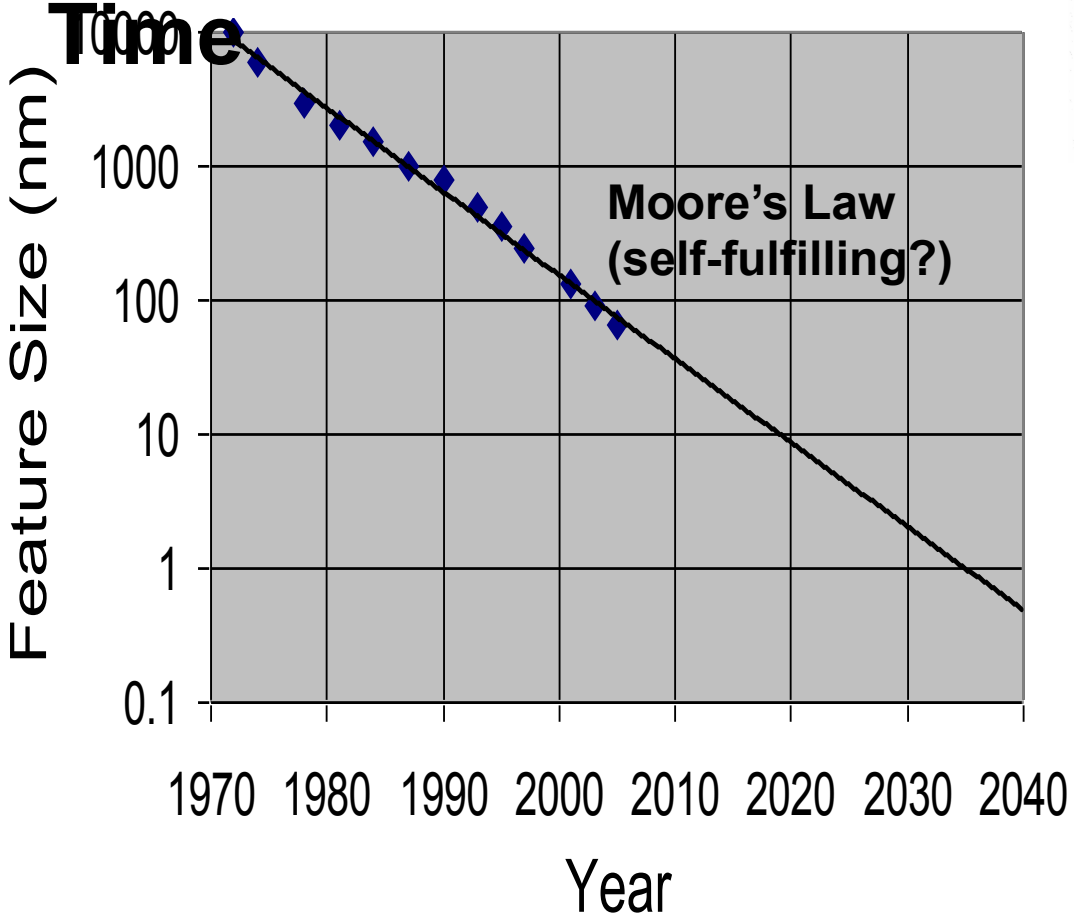


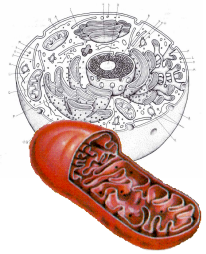


?

Into the Future

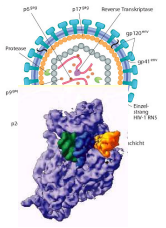
Computer Chip Feature Size versus Time





Eukaryotic cells

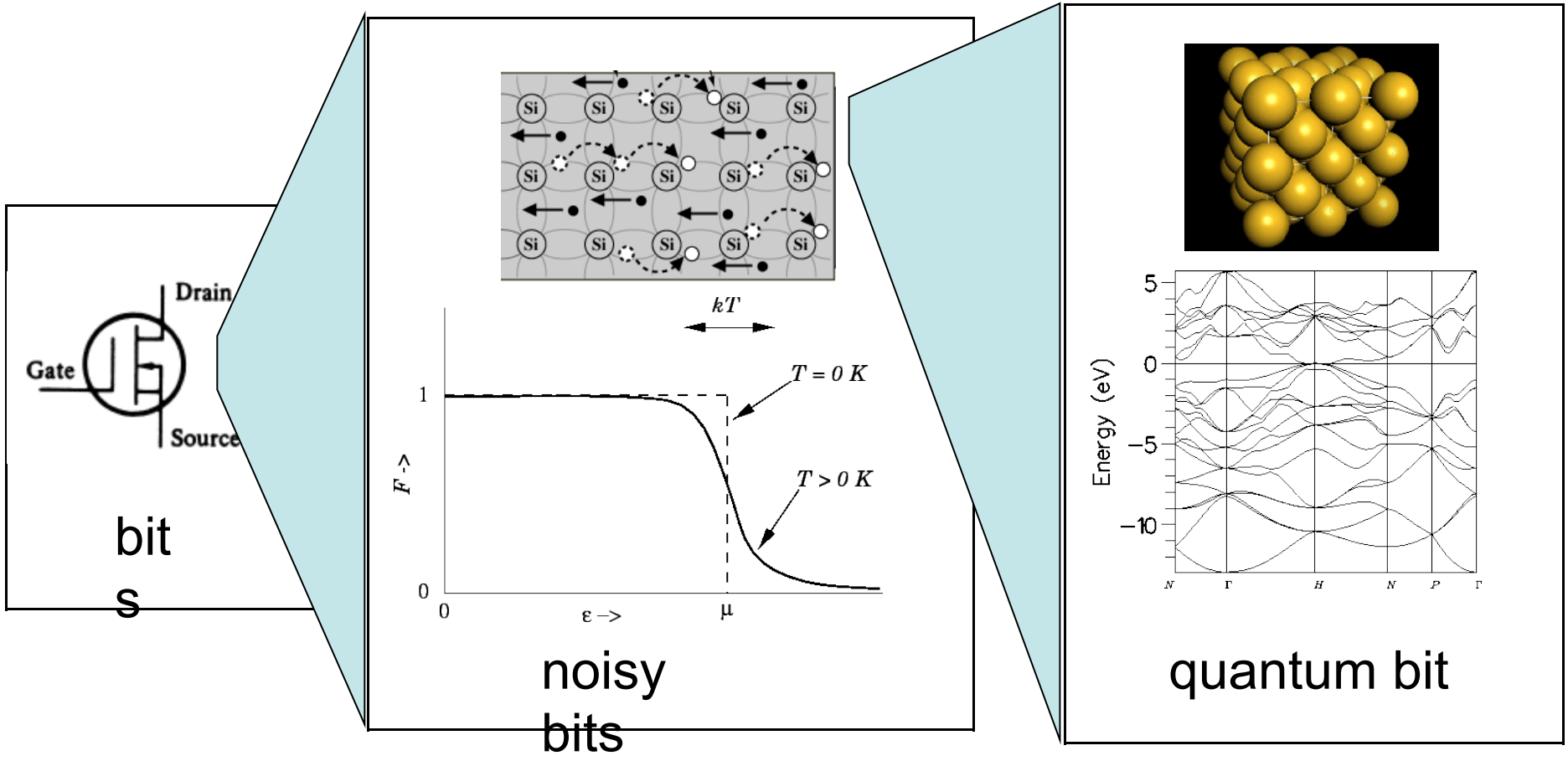
Mitochondria



AIDS virus

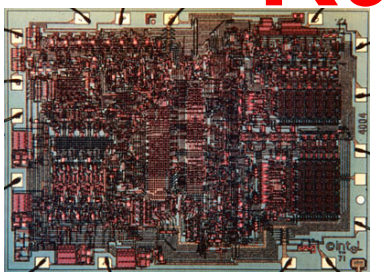
Amino acid

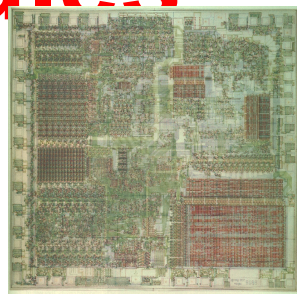
Behind the Bit

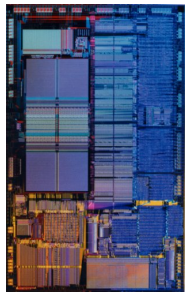


1. **Why is computation possible at all?**
2. **What does physics have to say about computation?**

New Sizes, New Rules



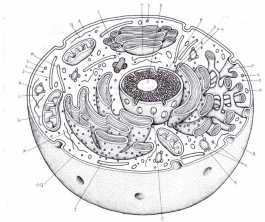




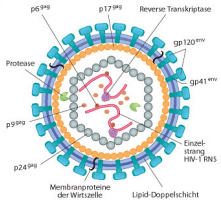
NOISY (FAULTY)

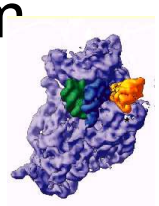
QUANTUM

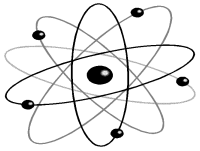
10,000nm 1,000nm 100nm 10nm 1nm 0.1nm







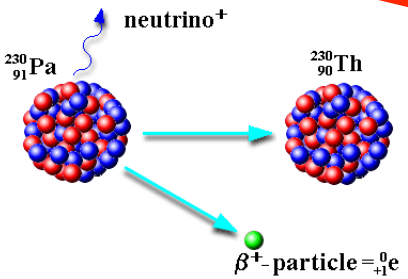




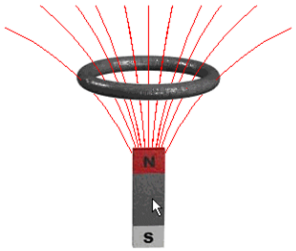
1. Can we build digital devices?
2. “Quantum” computers?

What is Quantum Theory?

Weak Force

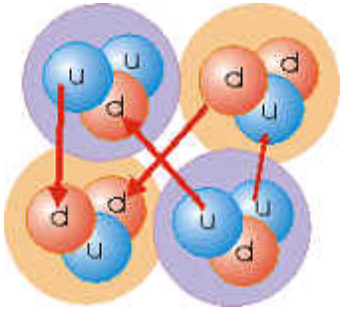


Electromagnetic Force

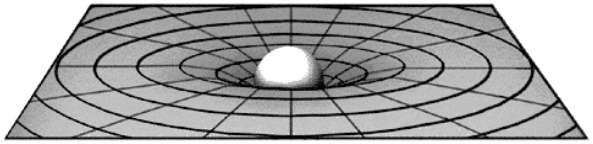




Strong Force



Gravitational Force (?!?!)



Quantum Theory

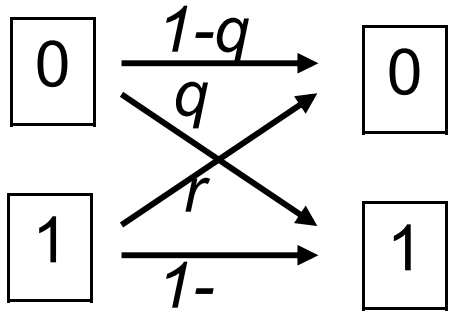
Quantum theory is the **"machine language"** of the universe

Probabilistic bits

Configuration: 0 1

Description: $\vec{p} = \begin{bmatrix} p \\ 1-p \end{bmatrix}$ 0 $p \in \mathbb{R}$, $0 \leq p \leq 1$; 1

Evolution:



$$M = \begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$$

Stochastic Matrix

$$\vec{p}_{out} = M\vec{p}_{in}$$

Measurement:

$$\vec{p} = \begin{bmatrix} p \\ 1-p \end{bmatrix} \rightarrow$$ 0 *p* percent; 1 *1-p* percent

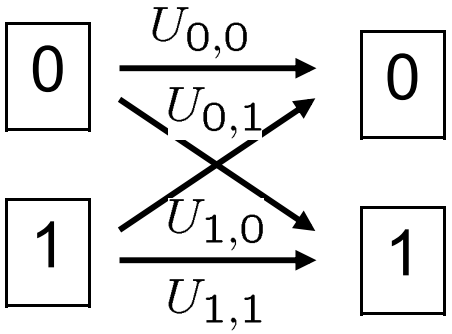
Quantum bits (qubits)

Configuration: 0 1

Description:

$$\vec{\psi} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix} \quad \alpha, \beta \in \mathbb{C} \quad |\alpha|^2 + |\beta|^2 = 1$$

Evolution:



$$M = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

Unitary Matrix

$$\vec{\psi}_{out} = M\vec{\psi}_{in}$$

Measurement:

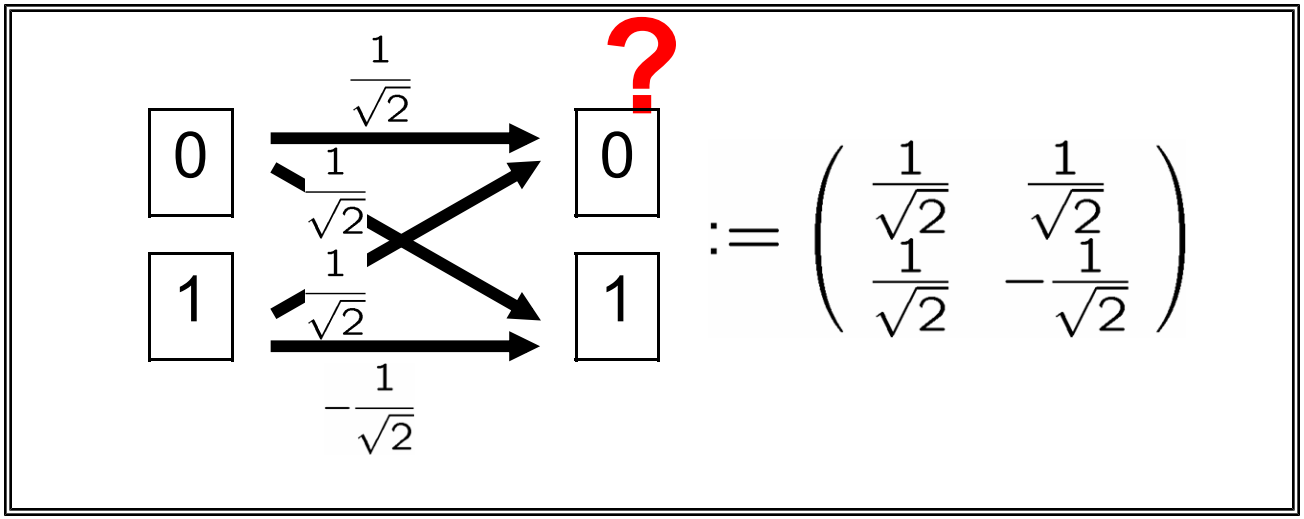
$$\vec{\psi} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{matrix} 0 \\ 1 \end{matrix} \quad \begin{matrix} |\alpha|^2 \text{percent} \\ |\beta|^2 \text{percent} \end{matrix}$$



Interference



?



qubit input

$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	100%
1	50 %	0%

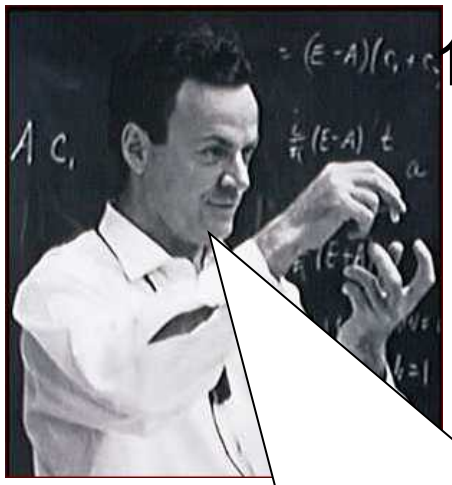
$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	0%
1	50 %	100%

A Worrying Physicist

2040 is a long time in the future….so why worry?

Richard Feynman



Observations

1. No efficient computer simulations of quantum theory
2. Systems evolving according to quantum theory efficiently compute properties of the quantum system (that's what experiments are!)

"How can we simulate quantum mechanics?... Can we do it with a new kind of computer…a **quantum computer**? It's not a Turing machine but a machine of a different kind" (1981)

Shocking Discoveries



Umesh Vazirani (U.C. Berkeley)

Quantum computers offer superpolynomial speedups (for certain oracle promise problems) over classical computers [Bernstein, Vazirani 93]

Quantum computers offer exponential speedups (for certain oracle promise problems) over classical computers [Simon 94]



Dan Simon (Microsoft)

The Factoring Firestorm

1881988129206079638386972394616 5043
9807163563379413827007633564229888
5971523466548531906060650474304 5317
3880113033967161996923212057340 3187
9550656996221305168759307650257059

=

39807508642406493739712550055038649119906436234252670840638518957594638895726176858331 7

×

47277214610743530253622307197304822463291469530209711645985217113052071125636359039752 7

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$



Peter Shor

Quantum computers can efficiently factor integers and can efficiently compute the discrete logarithm [Shor 94]

Quantum Computers!



Public Key Cryptosystems

RSA

BROKEN

Curves



188198812920607963838697239461650439807163563379417382700763356422988859715234654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570 59

Over the Top



“Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world’s most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security.” -Simon Singh

Quantum Computers!

1. The world obeys the laws of quantum theory
2. Computers obey the laws of physics
3. Computers should obey the laws of quantum theory
4. Quantum computers!

What can quantum computers do? Why do they work?

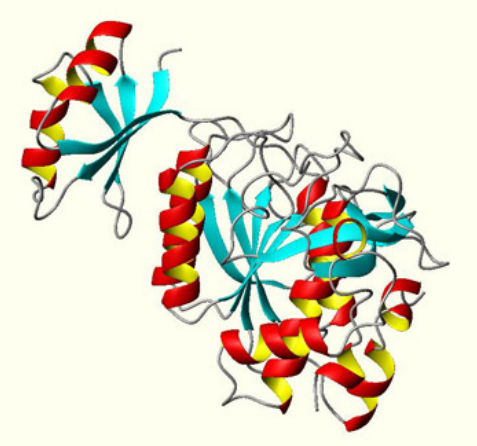
(Will Google buy one?)

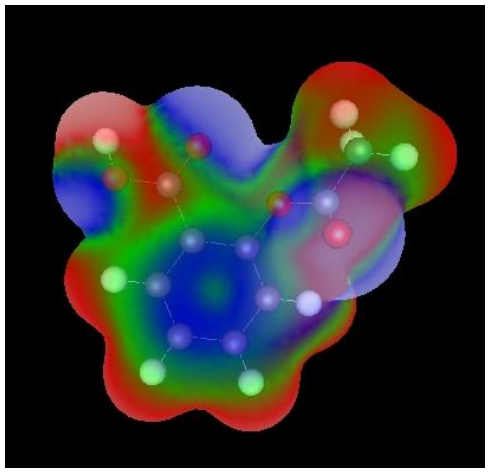
Can we actually build a quantum computer?

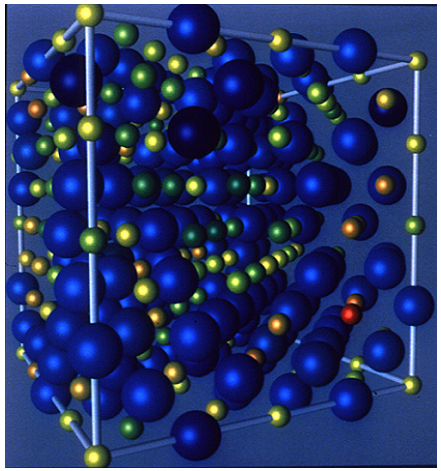
(Will Google buy one?)

What does the quantum model of computation tell us about the limits of computation in our universe?

Why Build a Quantum Computer? (if you're not the NSA)



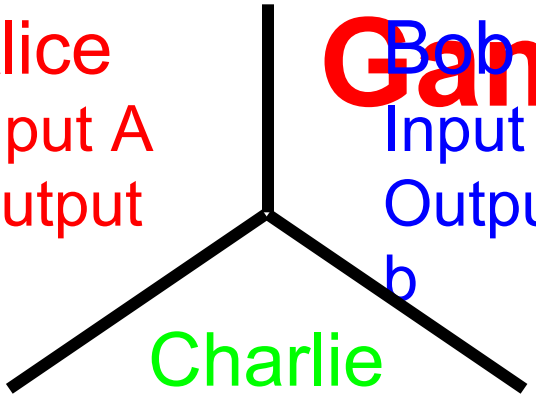




Simulation of quantum systems crucial to biology, chemistry, material science, ….

Quantum computers can do this! So why can they do this?

GHZ/Mermin Game



Input: X or Y, Output: +1 or -1

Winning Payoffs:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

deterministic local strategies, given by aX aY bX bY cX cY

aY bY cX = -1

aY bX cY = -1

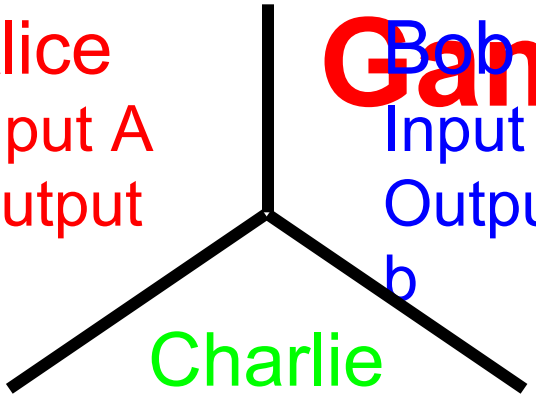
× aX bY cY = -1

aX bX cX = -1

→ not possible to always win → even with <u>prior</u> classical communication

convex sum of deterministic local stratgies

GHZ/Mermin Game



Input: X or Y, Output: +1 or -1

Winning Payoffs:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

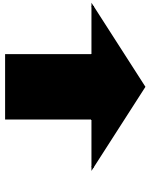
Quantum communication before game is played:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

Each party measure along X or Y direction

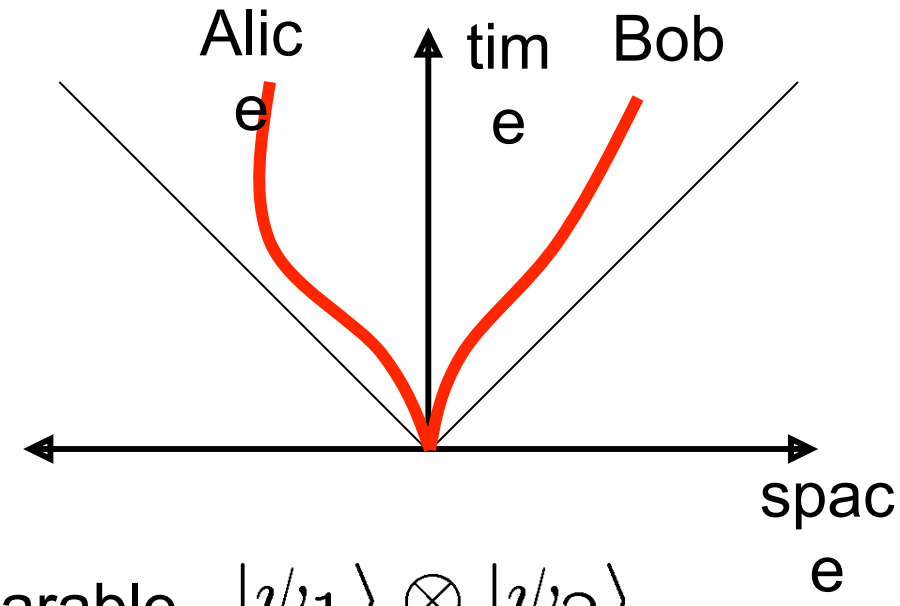
$\langle GHZ|XXX|GHZ\rangle = +1$

$\langle GHZ|YYX|GHZ\rangle = -1$



possible to always win!

Bell's Theorem



Separable $|\psi_1\rangle \otimes |\psi_2\rangle$

Entangled: $\neq |\psi_1\rangle \otimes |\psi_2\rangle$



John Bell

Measurements on space-like separated entangled quantum systems cannot be explained by local classical communication.

Minimizing the Yakking



ALICE

a

Feq(a,b) =

1 if a=b

0 if a≠b



BOB

b

0	1	0	1	•••	0

n bits

0	1	1	1	•••	0

n bits

How many bits do Alice and Bob need to exchange (in the worst case) to compute F(a,b)?

With no error: n+1 bits

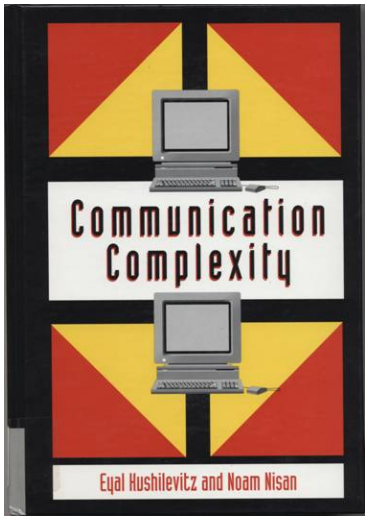
With one-sided error ε: O(log(n/ε)) bits

Communication Complexity



Classical communication costs of distributed classical computations

[Yao, STOC 79]

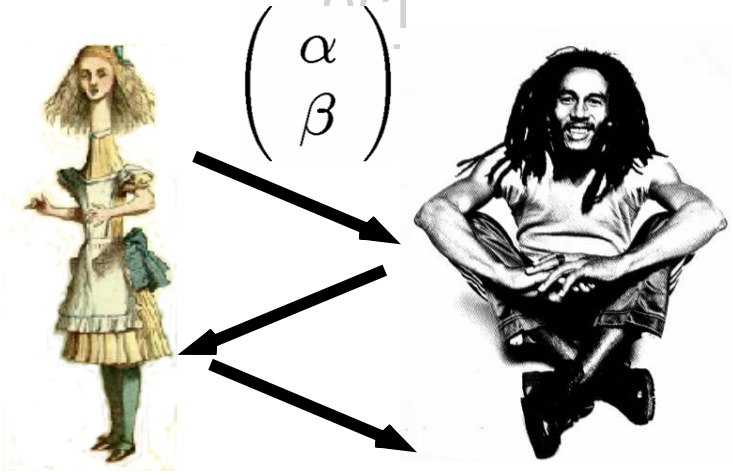


[Kushilevitz and Nisan



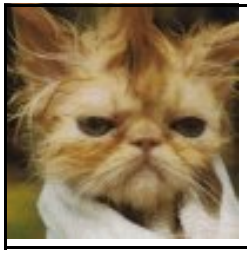
Quantum communication costs of distributed classical computations

[Yao, FOCS 93]



exchange qubits

Quantum Communication Complexity



Holevo's Theoerem: sending n classical bits over a quantum channel requires n qubits



[Raz, STOC 99]

There exist functions *F(a,b)* which require $\Omega(m1/4/\log(m))$ bits of communication to solve
(with error) but can be solved using only $O(\log(m))$ qubits of communication.
(*m* is the length of *a* and *b*)



Exponential benefit of quantum over classical.

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit

X

Y

2 bits





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ○ $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ ○ $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

Conditional on received bits, Bob performs manipulation of his half of entangled system

Rehas



[Raz, STOC 99]

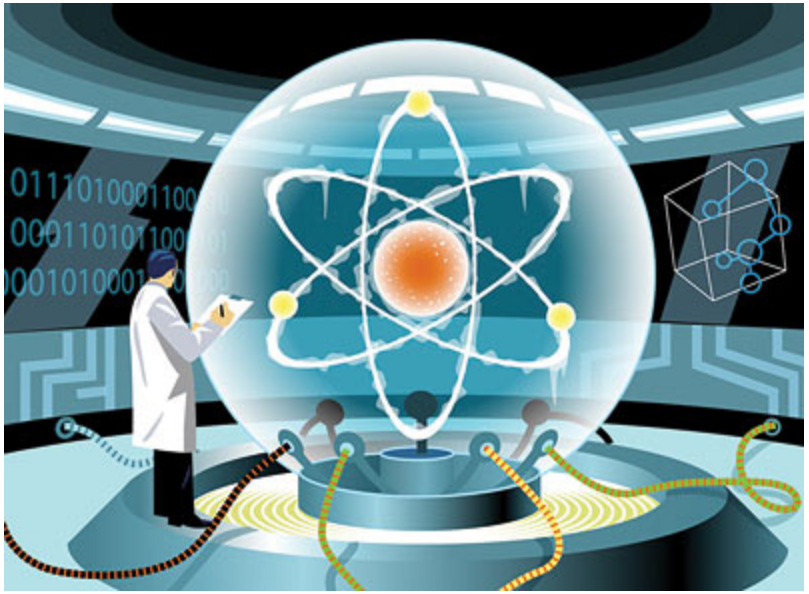
There exist functions *F(a,b)* which require *Ω(m1/4/log(m))* bits of communication to solve
(with error) but can be solved using only *O(log(m))* bits and *O(log(m))* ebits.
(*m* is the length of *a* and *b*)

1 ebit := $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$

1. Entanglement can be used to gain exponential reduction in communication resources
2. Entanglement is “like” classically correlated bits…but it’s different…

Preliminary thesis: entanglement is involved in quantum benefits

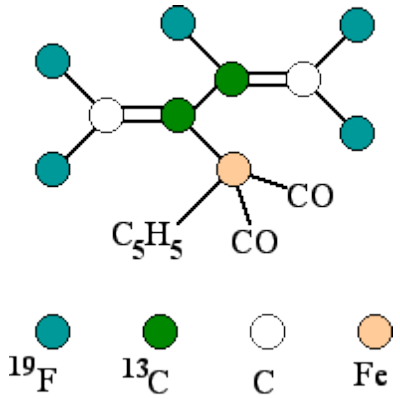
Entanglement Hungry



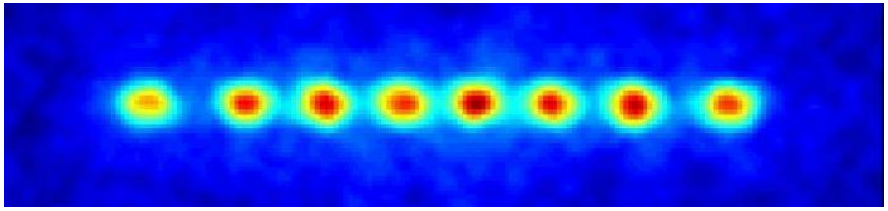
Entanglement is the resource consumed by quantum computers to achieve speedups over classical algorithms

Why aren’t quantum computers everywhere?

“Big” Quantum Computers



Isaac Chuang’s group
used 7 qubits to factor 15



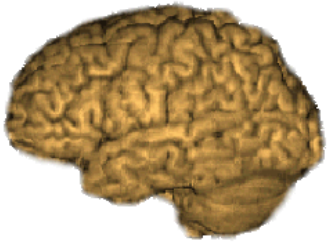
Rainer Blatt’s group entangled
a quantum byte (8 qubits)

from http://www.quantumoptics.at/

Two Views of Information

Information is "fungible" adj. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.







Al Gore

George Bush

Ralph Nader

Pat Buchanan

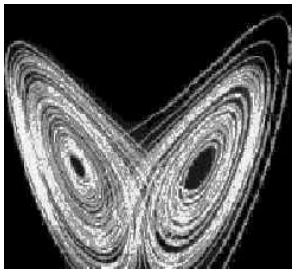




Rolf Landauer

No Information without Physical Representation!

Not all physical systems are robust information processing machines



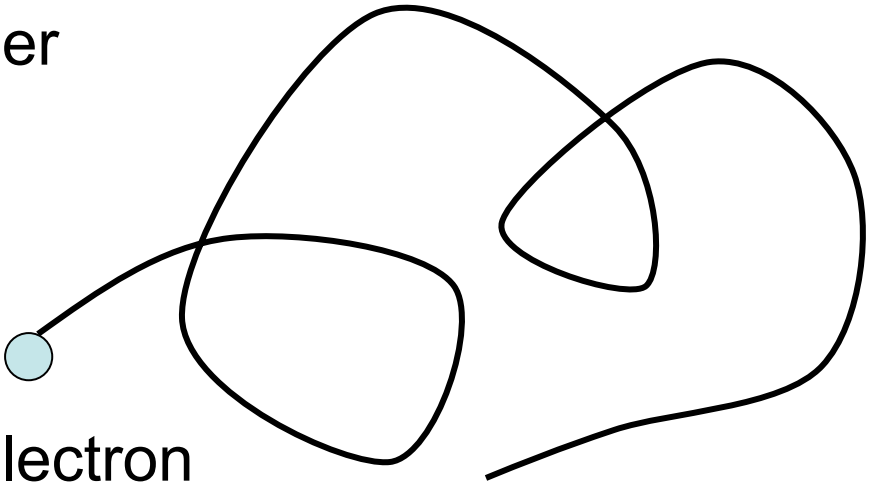
Information



Rolf Landauer

No Information without Physical Representation!

Not all physical systems are robust information processing machines



pobre electron

Position: (x,y,z)

Store information in position?

Analog Information?

$$\vec{\psi} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

two complex numbers, i.e. four real numbers

digits of precision?

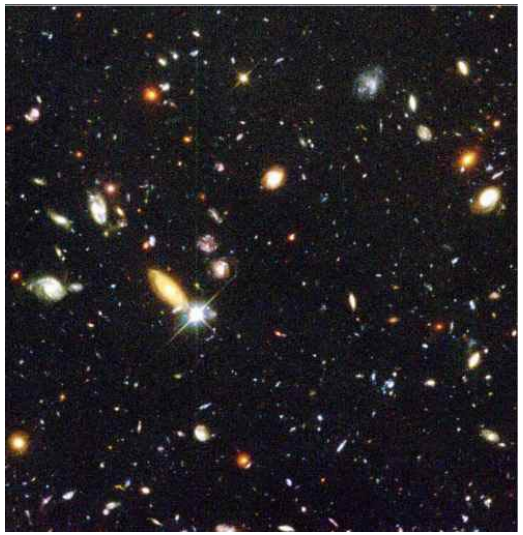
$$\vec{\psi} = \begin{bmatrix} 0.70710678118654752440084436210485\ldots \\ 0.70710678118654752440084436210485\ldots \end{bmatrix}$$

Worse yet, to describe quantum state of n qubits requires 2n complex numbers!

$$\vec{\psi} = [\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5 \ldots, \alpha_{2^n-1}]^T$$

300 qubits = 2300 ≈ 2×1090

Number of particles in universe ≈ 1072 - 1087

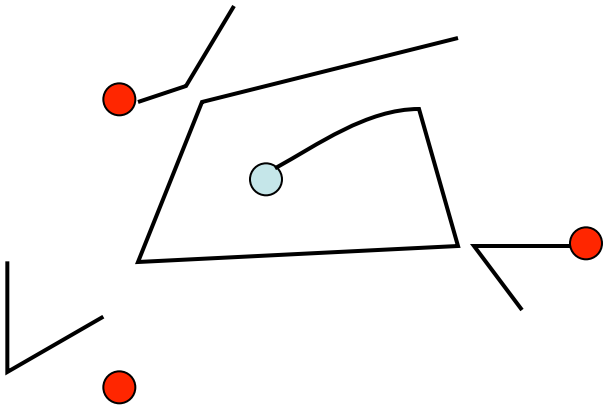


Noise

Position of particle:

$$x = 1.012318639234902349080 91\ldots$$

But particle is not isolated, so it's environment bumps it around



And measurement cannot be perfect

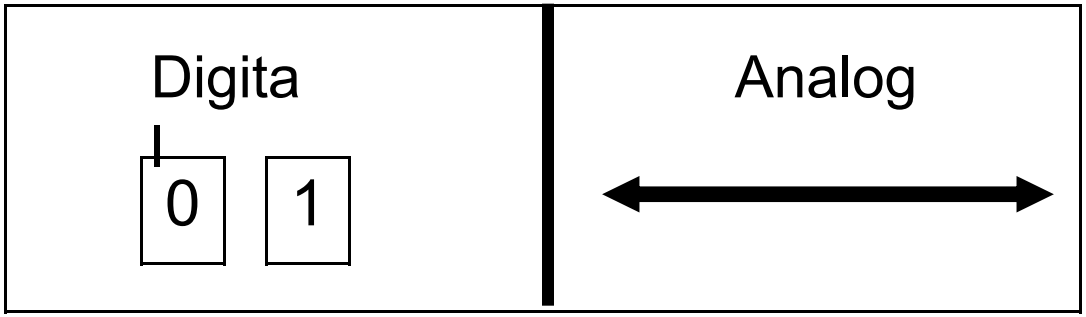
0	1	1	1	0	1	0	1	?



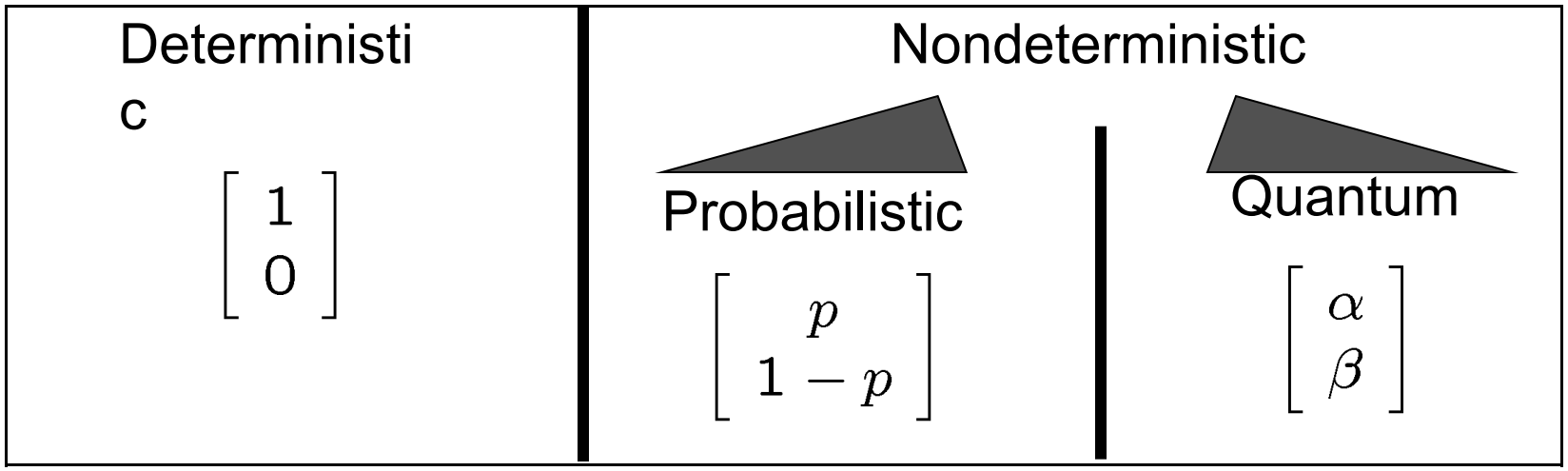
Each digit requires more
and more resources

A Trap

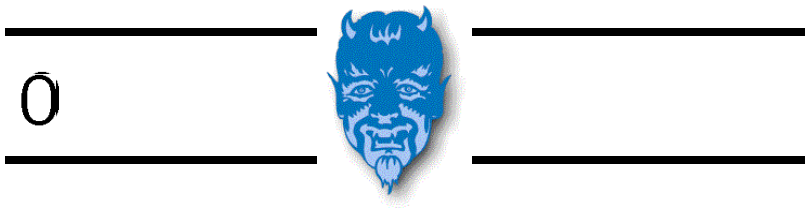
Common misconception



versus



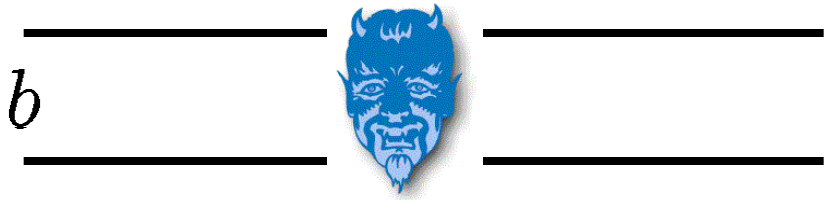
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix}000\\001\\010\\100\end{matrix}\right\} \rightarrow 0$$

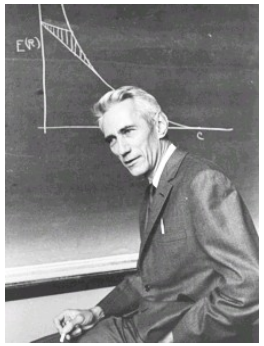
$$\left.\begin{matrix}111\\110\\101\\011\end{matrix}\right\} \rightarrow 1$$

Probability of failure =*3(1-p)p2+p3*

=3p2-2p3

<p if *p<1/2*

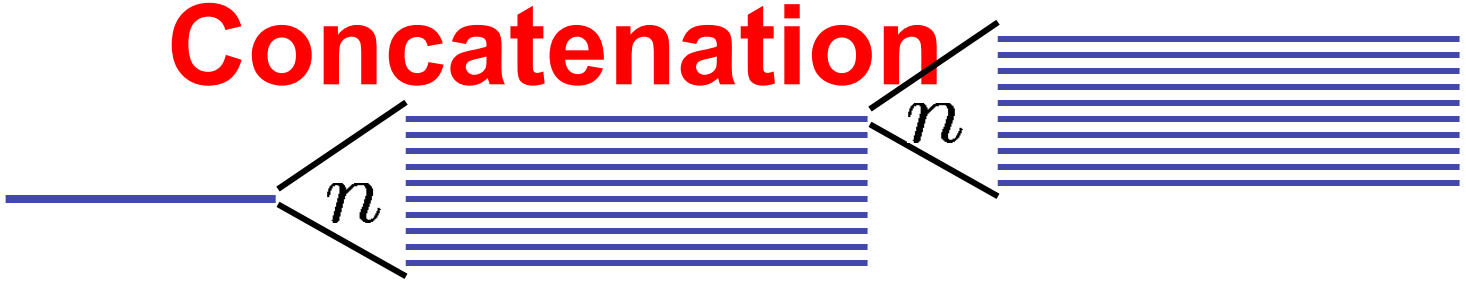
Use redundancy to protect information against noise



Claude Shannon 1948

“A mathematical theory of communication”

Perfection Through Concatenation



failure probability

p $\quad$ $\frac{cp}{2}$ $\quad$ $\frac{c(cp2)}{2}$

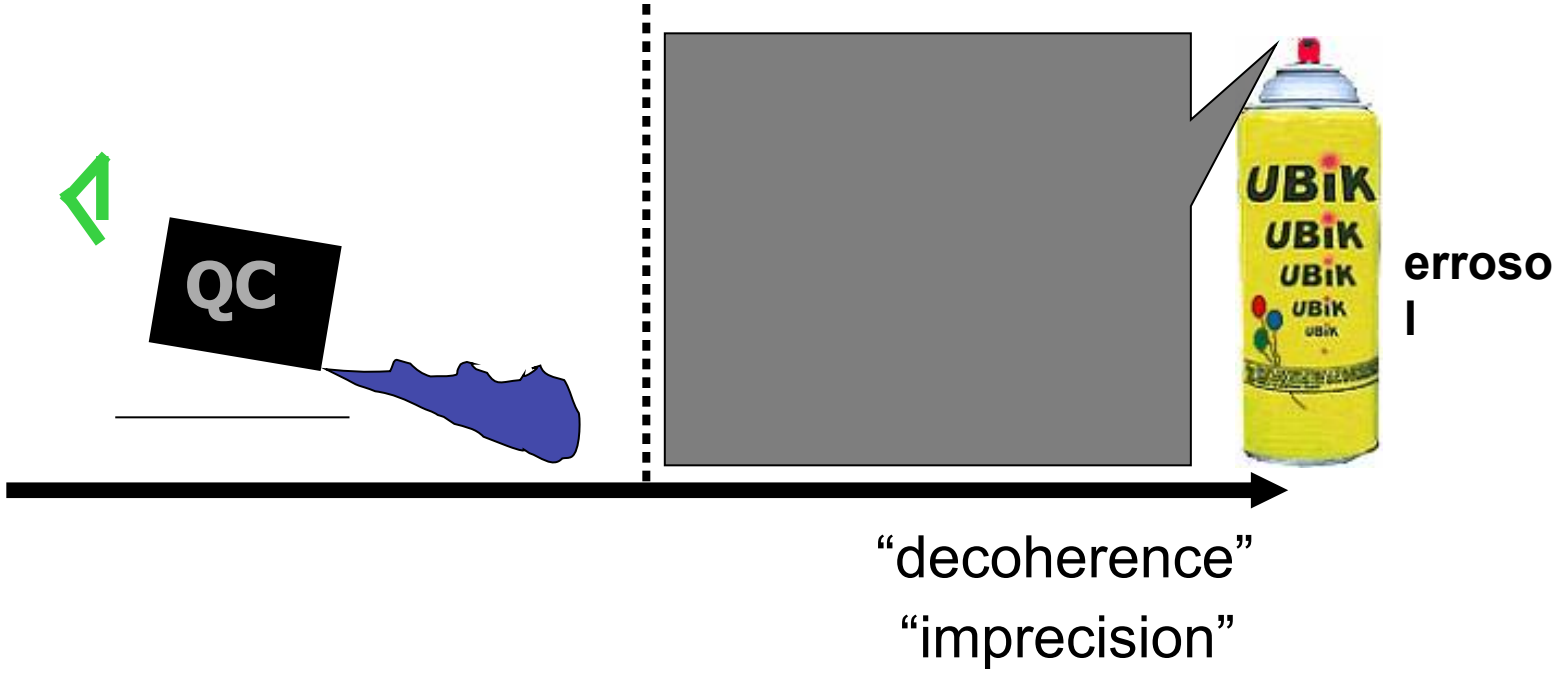
At kth level of concatenation, probability of an error which destroys the encoded information

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{bits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

Threshold Theorem, 1996



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Partial Answers

Why aren't quantum computers everywhere?

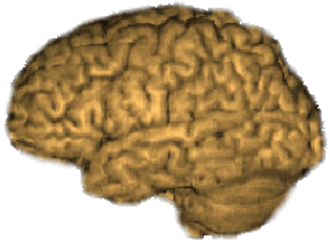


Noise can destroy quantum information

This can be overcome using complex quantum error correcting procedure... (Quantum computing is a valid model of computing!)

But ?





Whence Physics?

RANT MODE ON

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation

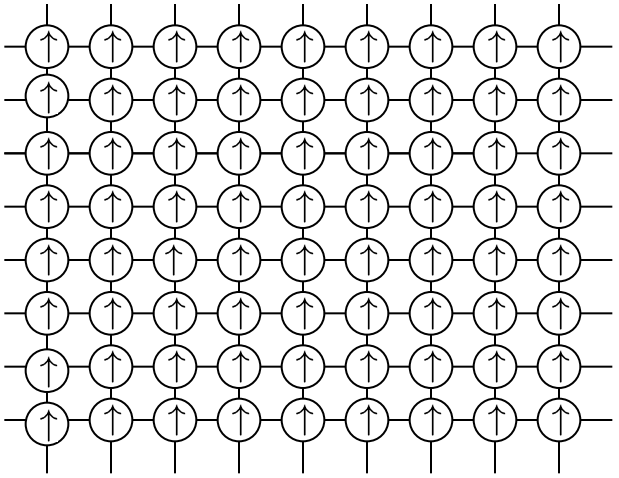
there exists systems whose <u>PHYSICS</u> guarantees their ability to enact robust classical computation.



THE BILLION DOLLAR QUANTUM QUESTION:

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

Toy Classical Model

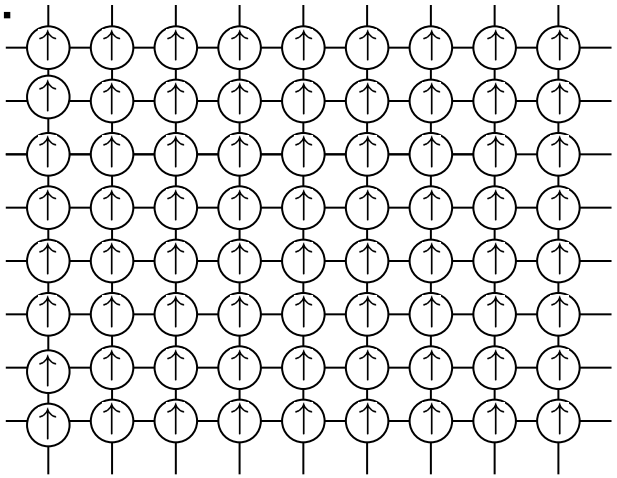


Ising model in two+ dimensions

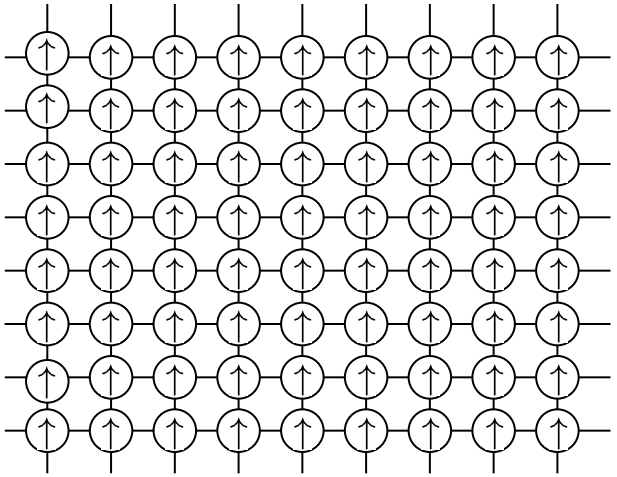
$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

Redundancy code:

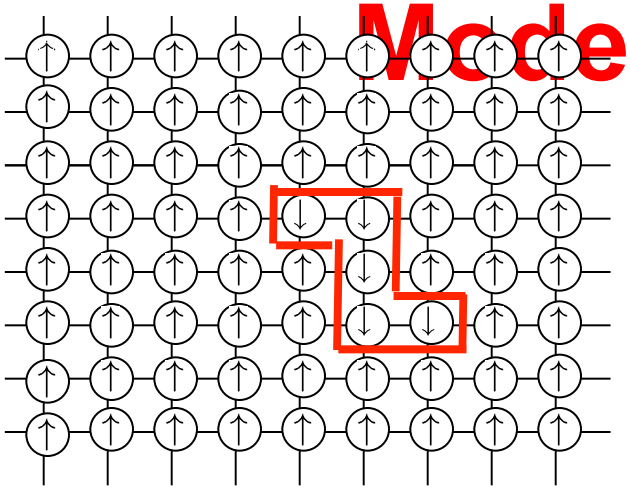
0 $\rightarrow$



1 $\rightarrow$

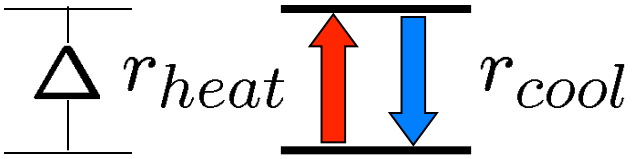


Dynamical Ising Model



Energy of excitation=
2J (Sum of perimeters of flipped domains)

Dynamics obeying principle of detailed balance:



$$\frac{r_{heat}}{r_{cool}} = e^{-\frac{\Delta}{k_B T}}$$

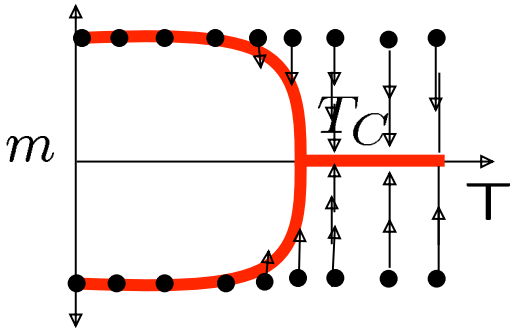
Competition between growth and decimation of flipped domains



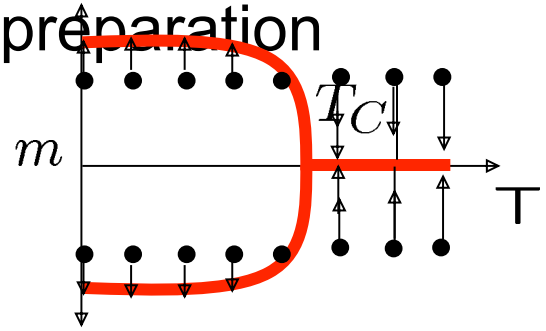
Robust



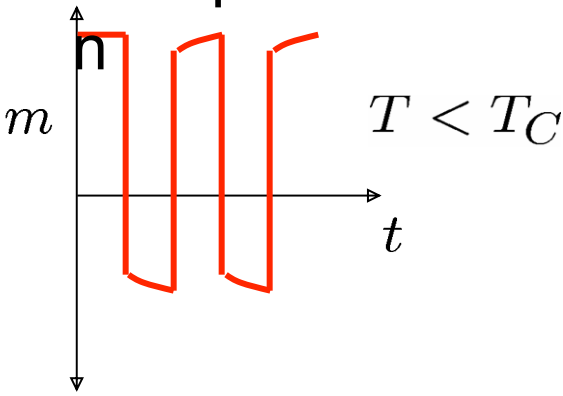
Relaxation to thermal



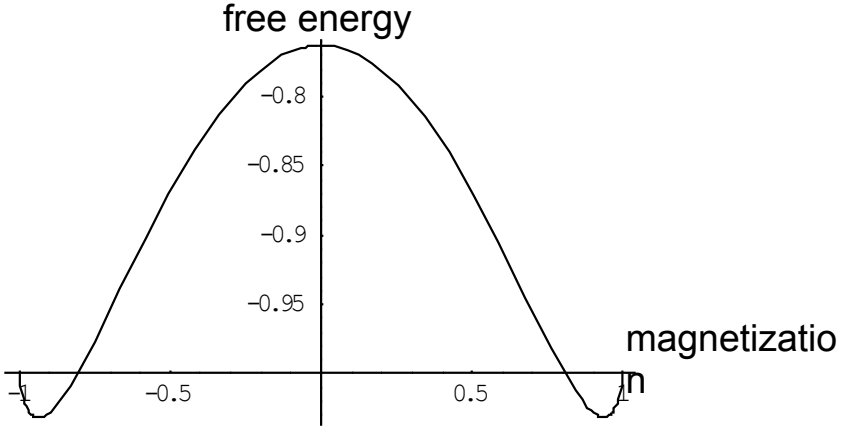
Imperfect preparation



Manipulation

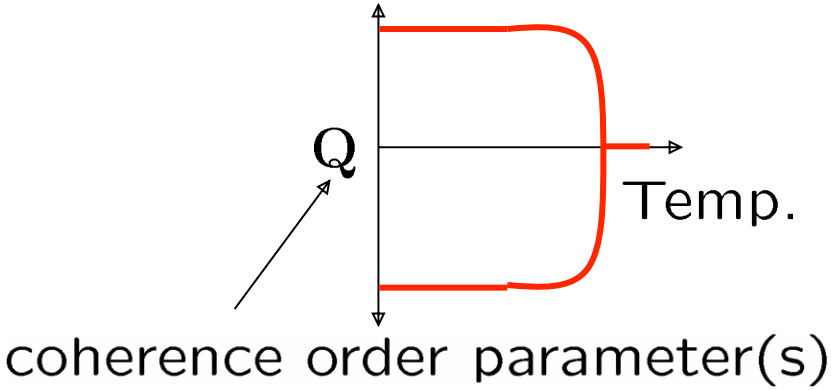


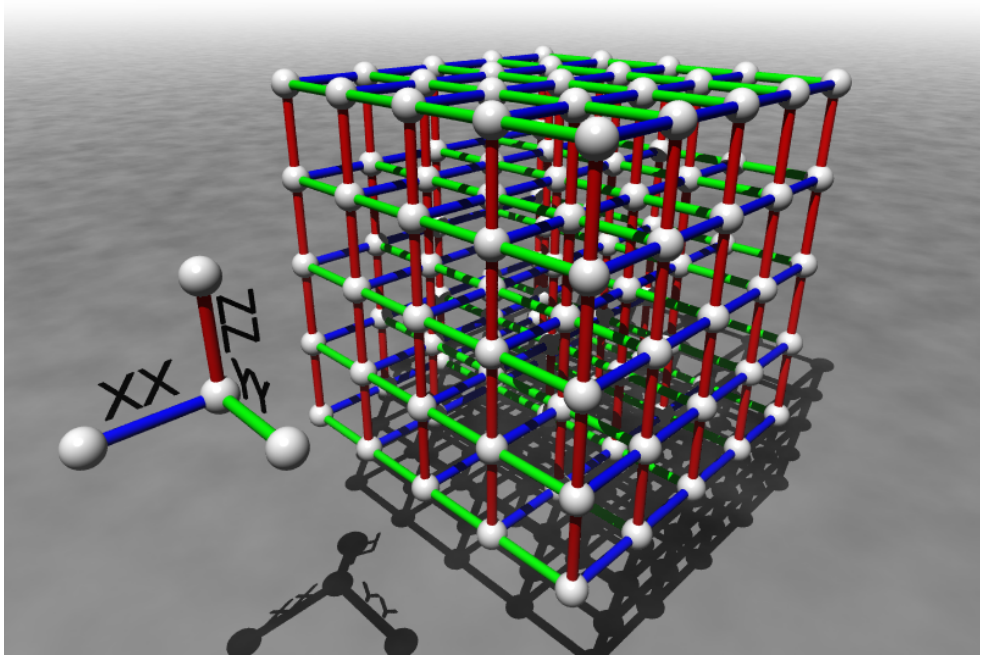
The two encoded states have lifetimes which grow with the size of the system:



Quantum Hard Drive?

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?





Into the Future

A **NEW** model of computation

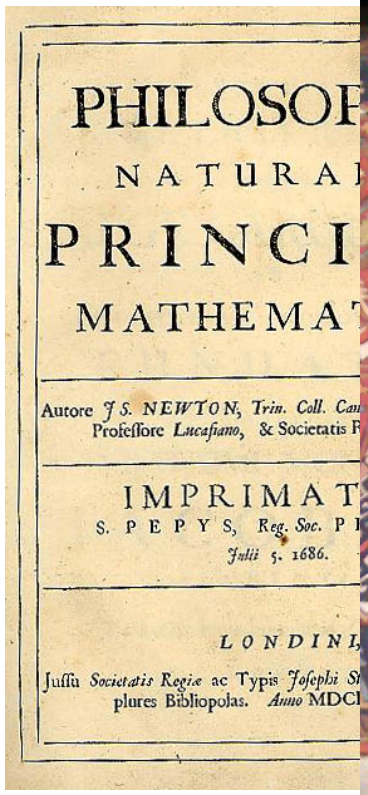
can we build it? what will it look like?

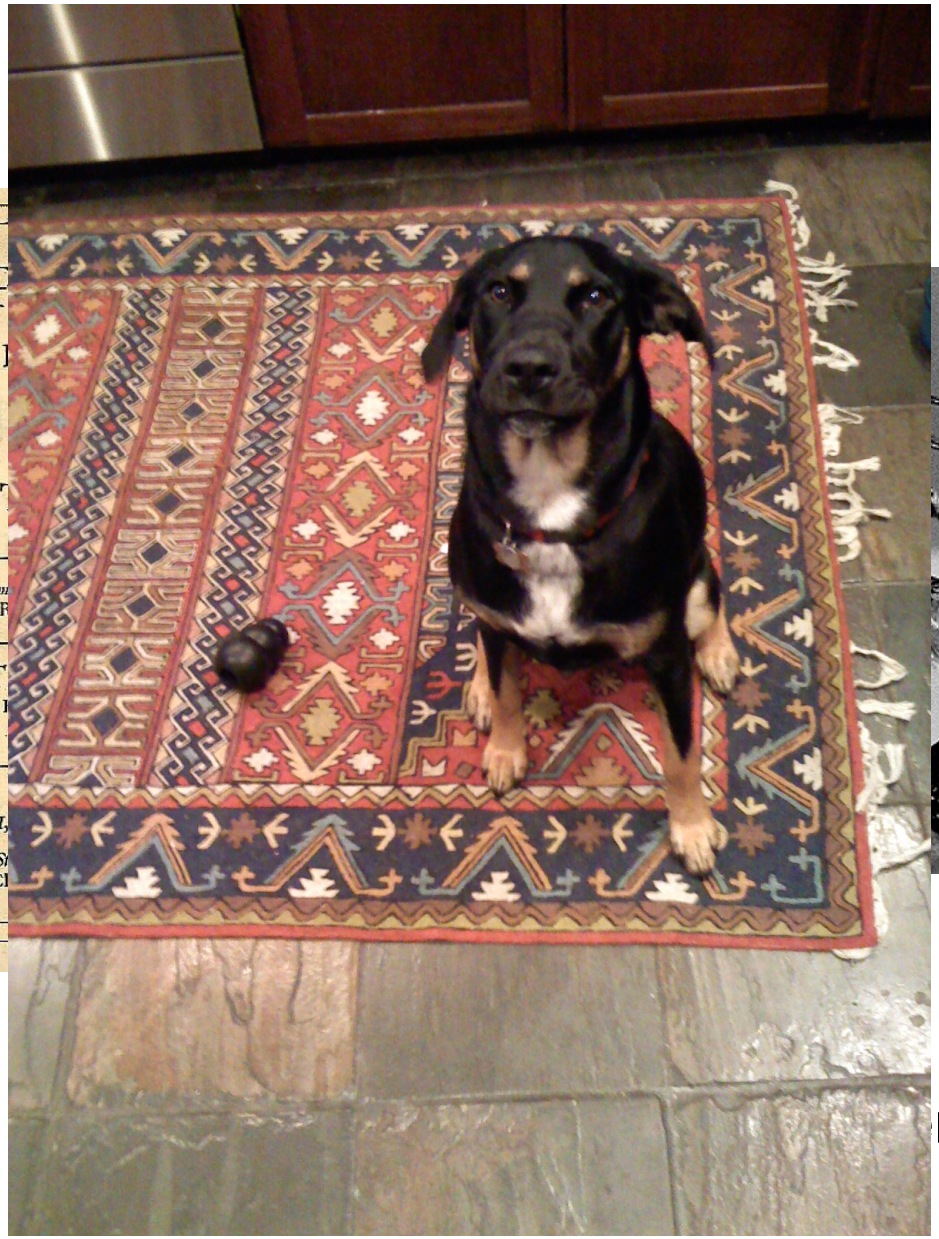
what ultimate computational powers does it have?

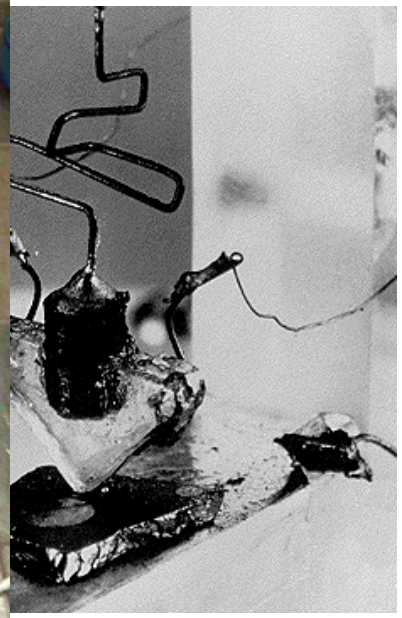
what ultimate computational powers does it lack?
[useful for physicists simulating quantum systems]

might it allow us to understand classical computation better?

What Gets Me Out of Bed


PHILOSOP
NATURAL
PRINCI
MATHEMAT
Autore J S. NEWTON, Trin. Coll. Cant
Professore Lucasiano, & Societatis R
IMPRIMAT
S. PEPYS, Reg. Soc. P
Julii 5. 1686.
LONDINI,
Jussu Societatis Regiæ ac Typis Josephi St
plures Bibliopolas. Anno MDCI





Intellectual
Merit

nces